

Twelve famous strategies, six markets, one set of rules: what we know, and what is missing

**In one paragraph.** Twelve strategies were run with the same code on six point-in-time universes (US, EU, UK, Denmark, Scandinavia, World), 2013–2026, after costs, each with a full factsheet: P&L, trade records, drawdowns, attribution, a pre-declared fix ladder, a classification and a 360° view. Only 1 of eleven long/short books (FS10) earns a **positive** return that passes the 2.87 gate in at least one market; 3 (FS01, FS06, FS12) pass it with a significantly **negative** return; long-only books beat the equal-weight universe's Sharpe most often in the momentum and low-risk families. Many signals are **redundant**: once their nearest library neighbours are in the model, 6 of eleven have no alpha left. The price-only battery collapses into a few roots: **momentum** (12-1, 52-week high, residual momentum), **low risk** (volatility, beta, MAX) and, weakly, **reversal** and **size**. The biggest gaps are data (fundamentals and delisted prices outside the US), models (no multivariate test yet) and implementation (flat trading costs and no market impact). Borrow fees, financing of leverage, World in one currency, a replication check and a programme-wide trial ledger were added on 27 September 2026 (§3.0).

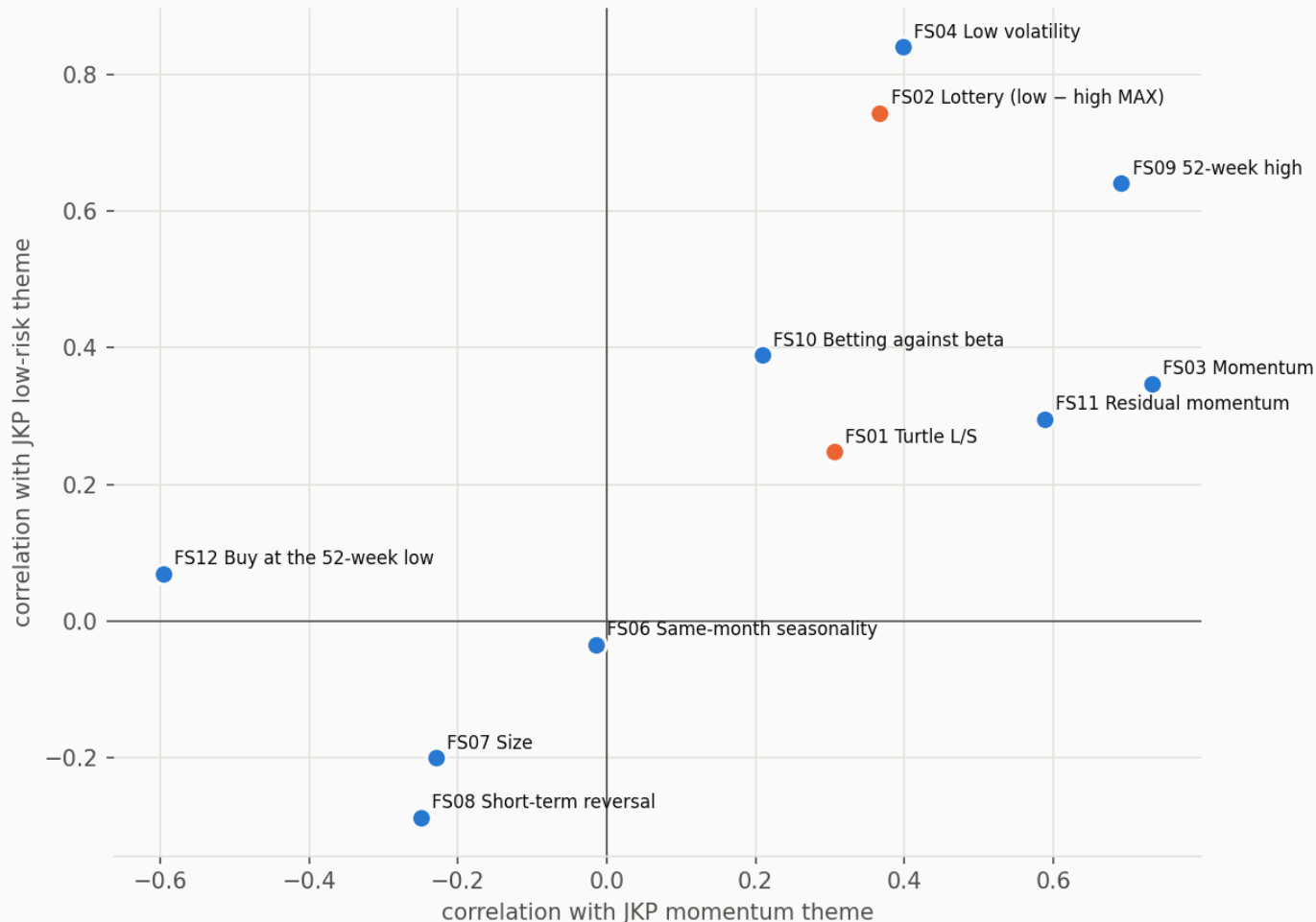
| Strategy | Family | L/S net return / yr (range over 6) | L/S positive | L/S passes |  $t$  |  $t > 2.87$  | L/S CAPM alpha range | Long-only beats EW Sharpe | Long-only alpha passes | Fixes: holdout Sharpe | Redundant in 360° view | Library cluster | |---|---|---|---|---|---|---|---| | FS01 Turtle Traders | Breakout & trend | -15.7% to -3.9% | 0 / 6 | US-, SCANDI-, World- | -13.8% to -2.4% | 0 / 6 | — | T0 -0.74 → T3 0.64 (drop shorts, 4N stop) | yes | momentum (time series) | | FS02 Lottery (MAX) | Lottery & attention | -14.4% to -1.6% | 0 / 6 | — | -5.5% to +4.7% | 2 / 6 | — | L0 -0.57 → L1 -0.41 (beta-neutral legs) | yes | low risk | | FS03 Momentum | Momentum & trend | -2.8% to +9.7% | 5 / 6 | — | +1.8% to +15.8% | 5 / 6 | UK, World | 0.29 → 0.55 (full ladder) | no | momentum | | FS04 Low volatility | Low risk | -15.3% to -1.6% | 0 / 6 | — | -1.6% to +8.4% | 5 / 6 | UK | -0.27 → 0.18 (full ladder) | no | low risk | | FS05 Monkey portfolios | Portfolio folklore | beat the index: 16–86% of monkeys | — | — | — | — | — | no fix: a benchmark | — | size (equal weight) | | FS06 Same-month seasonality | Calendar & seasonality | -12.7% to +3.1% | 1 / 6 | EU- | -12.5% to +2.6% | 1 / 6 | — | -0.62 → -0.56 (full ladder) | yes | momentum | | FS07 Size | Size | -3.0% to +4.9% | 4 / 6 | — | -5.5% to +2.9% | 1 / 6 | UK | -0.44 → -0.45 (full ladder) | yes | momentum | | FS08 Short-term reversal | Reversal & bottom-fishing | -9.7% to +0.1% | 1 / 6 | — | -13.1% to -1.0% | 0 / 6 | — | -0.91 → -0.87 (full ladder) | no | momentum | | FS09 52-week high | Momentum & trend | -5.9% to +5.8% | 2 / 6 | — | +5.3% to +15.2% | 5 / 6 | EU, DK | 0.02 → 0.59 (full ladder) | no | momentum | | FS10 Betting against beta | Low risk | +2.0% to +9.5% | 6 / 6 | UK+ | -0.8% to +7.9% | 5 / 6 | UK | 0.00 → 0.07 (full ladder) | no | low risk | | FS11 Residual momentum | Momentum & trend | +1.9% to +7.9% | 6 / 6 | — | +4.3% to +10.7% | 6 / 6 | — | 0.52 → 0.57 (full ladder) | yes | momentum | | FS12 Buy at the 52-week low | Reversal & bottom-fishing | -17.0% to -5.2% | 0 / 6 | US-, EU-, World- | -16.6% to -6.4% | 0 / 6 | — | -0.96 → -0.78 (full ladder) | yes | momentum |

Net of costs, local currency, Feb 2013 – Aug 2026 (FS01 daily from Jan 2013). In the gate column + marks a significantly positive return, – a significantly negative one. The gate is Bonferroni over 12 cells per factsheet; fix ladders are judged on the 2020–26 holdout. Redundant =  $\alpha |t| < 2$  in every universe after the signal's closest library neighbours and the market.

#### Eleven strategies, six markets, 2013–2026 (FS05 monkeys are shown separately)

|                             | Long/short Sharpe ratio (net) |       |       |       |        |       |                             | Long-only Sharpe minus equal-weight universe Sharpe |       |       |       |        |       |
|-----------------------------|-------------------------------|-------|-------|-------|--------|-------|-----------------------------|-----------------------------------------------------|-------|-------|-------|--------|-------|
|                             | US                            | EU    | UK    | DK    | SCANDI | World |                             | US                                                  | EU    | UK    | DK    | SCANDI | World |
| FS01 Turtle Traders         | -0.86                         | -0.47 | -0.53 | -0.25 | -0.83  | -0.79 | FS01 Turtle Traders         | -0.74                                               | -0.39 | -0.29 | -0.62 | -0.74  | -0.44 |
| FS02 Lottery (MAX)          | -0.81                         | -0.11 | -0.23 | -0.28 | -0.18  | -0.69 | FS02 Lottery (MAX)          | -0.16                                               | +0.07 | +0.21 | -0.21 | -0.17  | -0.10 |
| FS03 Momentum               | +0.32                         | +0.34 | +0.43 | +0.31 | -0.19  | +0.41 | FS03 Momentum               | +0.12                                               | +0.18 | +0.41 | +0.23 | -0.12  | +0.30 |
| FS04 Low volatility         | -0.60                         | -0.08 | -0.30 | -0.08 | -0.26  | -0.48 | FS04 Low volatility         | -0.07                                               | +0.24 | +0.28 | +0.16 | +0.12  | +0.09 |
| FS06 Same-month seasonality | -0.67                         | -0.83 | +0.22 | -0.85 | -0.60  | -0.30 | FS06 Same-month seasonality | -0.30                                               | -0.36 | +0.05 | -0.44 | -0.21  | -0.09 |
| FS07 Size                   | +0.04                         | -0.18 | +0.45 | -0.21 | +0.25  | +0.02 | FS07 Size                   | -0.18                                               | -0.21 | +0.32 | -0.18 | -0.00  | -0.10 |
| FS08 Short-term reversal    | -0.36                         | -0.61 | -0.17 | -0.38 | +0.01  | -0.49 | FS08 Short-term reversal    | -0.42                                               | -0.42 | -0.30 | -0.25 | -0.14  | -0.36 |
| FS09 52-week high           | -0.28                         | +0.20 | -0.03 | +0.33 | -0.10  | -0.20 | FS09 52-week high           | -0.16                                               | +0.43 | +0.19 | +0.27 | +0.14  | +0.00 |
| FS10 Betting against beta   | +0.12                         | +0.34 | +0.81 | +0.17 | +0.21  | +0.39 | FS10 Betting against beta   | -0.01                                               | +0.14 | +0.33 | +0.09 | +0.11  | +0.10 |
| FS11 Residual momentum      | +0.20                         | +0.25 | +0.14 | +0.46 | +0.18  | +0.18 | FS11 Residual momentum      | +0.11                                               | +0.24 | +0.10 | +0.08 | +0.12  | +0.12 |
| FS12 Buy at the 52-week low | -0.87                         | -0.96 | -0.67 | -0.43 | -0.36  | -0.91 | FS12 Buy at the 52-week low | -0.43                                               | -0.62 | -0.21 | -0.22 | -0.33  | -0.38 |

#### All factsheets in factor space (long/short books; US, UK, DK, World average)



## 2. What the twelve factsheets say together

1. **Two roots carry almost everything.** The momentum family (FS03, FS09, FS11, and the breakout inside FS01) and the low-risk family (FS02, FS04, FS10) are where the beta-adjusted alphas are. Within each family the signals are close substitutes.
2. **Raw long/short numbers mislead in a bull market.** The long/short books of FS03, FS04, FS09 carry market betas of -1.00 to -0.29 against the equal-weight universe; in a rising market that short beta alone cost 4–14% a year (beta  $\times$  universe return). Beta-neutral legs remove most of this drag, but the neutralised books rarely pass the gate.
3. **Costs matter most for the fast signals.** Trading costs take 3.9–4.9% a year from short-term reversal (FS08) and 4.0–5.0% from seasonality (FS06), against 1.7–2.4% for 12-1 momentum and 1.0–1.4% for low volatility; both fast signals are negative after costs in most markets.
4. **Folklore fails where it contradicts momentum.** Buying at the 52-week low (FS12) and buying breakouts with tight stops on single stocks (FS01) both lose; the monkeys (FS05) only reflect whether small beat big.
5. **Survivorship was worth 2–5 percentage points a year** on the equal-weight universes outside the US; moving to point-in-time membership changed several conclusions (FS02).
6. **Six markets are not six tests.** The long/short books co-move strongly across universes (World contains the US; EU contains the Nordic blue chips), so the six markets are worth only 1.5–1.9 independent tests (effective number = 36 / sum of the 6 $\times$ 6 correlation matrix). "Positive in all six" is weaker evidence than it sounds.

| Strategy (L/S net)          | Mean pairwise correlation | US–World | EU–SCANDI | Effective number of markets |
|-----------------------------|---------------------------|----------|-----------|-----------------------------|
| FS03 Momentum               | 0.53                      | 0.85     | 0.58      | 1.6                         |
| FS04 Low volatility         | 0.60                      | 0.91     | 0.62      | 1.5                         |
| FS09 52-week high           | 0.62                      | 0.89     | 0.60      | 1.5                         |
| FS10 Betting against beta   | 0.57                      | 0.90     | 0.71      | 1.6                         |
| FS11 Residual momentum      | 0.44                      | 0.84     | 0.50      | 1.9                         |
| FS12 Buy at the 52-week low | 0.45                      | 0.85     | 0.62      | 1.9                         |

1. **Across the whole programme, few winners survive.** The factsheet trial ledger holds 205 gated tests. Under a programme-wide Benjamini–Hochberg correction 9 positive results survive (3 also Bonferroni,  $|t| > 3.67$ ) against 17 reliable losers; the best positive candidates have a deflated Sharpe ratio of 0.79 at most even on the lenient bound, below the usual 0.95 (planning/FACTSHEET\_TRIAL\_LEDGER.md).

### 3. Gap analysis

| Factor coverage today: stock-level in the six universes |             |             |             |             |             |             |
|---------------------------------------------------------|-------------|-------------|-------------|-------------|-------------|-------------|
| Momentum & trend                                        | covered     | covered     | covered     | covered     | covered     | covered     |
| Short-term reversal                                     | covered     | covered     | covered     | covered     | covered     | covered     |
| Low risk (vol, beta, MAX)                               | covered     | covered     | covered     | covered     | covered     | covered     |
| Seasonality                                             | covered     | covered     | covered     | covered     | covered     | covered     |
| Size                                                    | covered     | partial     | partial     | partial     | partial     | partial     |
| Value                                                   | partial     | not covered | partial     | partial     | not covered | partial     |
| Quality / profitability                                 | partial     | not covered | partial     | partial     | not covered | partial     |
| Investment / asset growth                               | partial     | not covered | partial     | partial     | not covered | partial     |
| Earnings momentum / PEAD                                | partial     | not covered | partial     | partial     | not covered | partial     |
| Analyst revisions                                       | not covered | not covered | not covered | not covered | not covered | not covered |
| Dividend / payout                                       | partial     | not covered | partial     | partial     | not covered | partial     |
| Liquidity                                               | partial     | partial     | partial     | partial     | partial     | partial     |
| Short interest                                          | not covered | not covered | not covered | not covered | not covered | not covered |
| News / sentiment                                        | not covered | not covered | not covered | not covered | not covered | not covered |
| Option-implied risk                                     | not covered | not covered | not covered | not covered | not covered | not covered |
|                                                         | US          | EU          | UK          | DK          | SCANDI      | World       |

partial = stock level in the US only (Sharadar fundamentals, not yet built) or factor-level returns elsewhere (JKP: UK, DK, World; none for EU, SCANDI);  
liquidity = traded-value proxy only

#### 3.0 Closed since the first edition (27 Sep 2026)

| Category       | Gap                                                                               | What was done                                                                                                                                                                                                |
|----------------|-----------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Data           | No FX conversion;<br>World sums local-currency returns                            | World converted to USD, unhedged, with Fed H.10 daily rates (EUR, GBP, DKK, SEK) and Yahoo PLN from 2015; the 21 Polish names stay in PLN before 2015 ( <code>data.to_usd</code> ).                          |
| Model          | Programme-level multiple testing not yet consolidated                             | One ledger of every gated test in FS01–FS13 with programme-wide Bonferroni, Benjamini–Hochberg and deflated Sharpe ( <code>code/ledger.py</code> , <code>planning/FACTSHEET_TRIAL_LEDGER.md</code> ).        |
| Implementation | Short side: no borrow fees or availability in the sorts (50 bp flat in FS01 only) | Every short leg pays a size-tiered borrow fee: 0.25% a year for the largest half of the universe, 0.75% for the next 30%, 2% for the smallest 20% ( <code>code/borrow.py</code> ); FS01 keeps its flat 0.5%. |

| Category       | Gap                                                                         | What was done                                                                                                                                                                                                                 |
|----------------|-----------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Implementation | Leverage without financing: BAB (rf = 0, no spread), vol targeting up to 3× | Net long cash in beta-neutral and BAB books is charged at the USD risk-free rate + 0.5% (borrow. financing). Volatility targeting scales a dollar-neutral book and is not charged for cash; its margin cost is still missing. |
| Validation     | No replication check against published factor returns                       | FS13: our US signals correlate 0.8 or more with the matching JKP factor for 8 of 12.                                                                                                                                          |
| Process        | Factsheet trials outside the programme ledger                               | All factsheet trials are now in the factsheet trial ledger; FS14 will be pre-registered there.                                                                                                                                |

### 3.1 Priority gaps still open (impact high, effort lowest first)

| Category       | Gap                                                                                                                         | Impact | Effort                      | Remedy                                                                                                                              |
|----------------|-----------------------------------------------------------------------------------------------------------------------------|--------|-----------------------------|-------------------------------------------------------------------------------------------------------------------------------------|
| Validation     | The six universes overlap and co-move: World contains US, UK and EU; EU contains the Danish, Swedish and Finnish blue chips | High   | Low                         | Report the effective number of markets; judge on pooled evidence; treat World as a summary, not a seventh market; run EU ex-Nordics |
| Data           | Residual survivorship outside the US: 9–25% of index member-quarters have no price (mostly delisted names)                  | High   | Medium                      | Buy delisted-inclusive history for Europe, or lengthen with the ART archive (1996–2013)                                             |
| Model          | Sort-based tests only; no multivariate (Fama-MacBeth) regressions                                                           | High   | Medium                      | Monthly cross-sectional regressions on all library signals, per universe                                                            |
| Implementation | Flat 10 bp per side; no size- or liquidity-dependent costs, no market impact                                                | High   | Medium                      | Spread and impact model by traded value (e.g. square-root impact)                                                                   |
| Coverage       | Missing factor families: value, quality, investment, earnings momentum, analyst revisions, payout, liquidity                | High   | Medium (US) / High (non-US) | US stock-level from Sharadar now; JKP factor-level for UK, DK, World                                                                |
| Data           | No company fundamentals outside the US                                                                                      | High   | High                        | JKP factor-level returns for UK, DK, World now; a licensed global fundamentals feed later                                           |
| Data           | Oslo missing from SCANDI; DK has ~19 names                                                                                  | Medium | Low                         | Add OBX history; report DK only as a robustness market                                                                              |
| Model          | Raw 252-day betas, no shrinkage                                                                                             | Medium | Low                         | Vasicek or Frazzini-Pedersen shrinkage                                                                                              |

### 3.2 Data

| Gap                                                                                                        | Where it bites                                                                       | Impact | Effort | Remedy                                                                                    | Source we already have                                                                  |
|------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------|--------|--------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------|
| Residual survivorship outside the US: 9–25% of index member-quarters have no price (mostly delisted names) | all non-US results; short legs penalised, long legs flattered                        | High   | Medium | Buy delisted-inclusive history for Europe, or lengthen with the ART archive (1996–2013)   | ART PIT database (Project1, to 2013); none for 2013–26                                  |
| No company fundamentals outside the US                                                                     | value, quality, investment, payout and earnings families missing in 5 of 6 universes | High   | High   | JKP factor-level returns for UK, DK, World now; a licensed global fundamentals feed later | JKP 153 factors per country (Project2 cache); Sharadar US fundamentals (staged, unused) |

| Gap                                                                                                   | Where it bites                                                                             | Impact | Effort      | Remedy                                                   | Source we already have                                                                |
|-------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|-------------|----------------------------------------------------------|---------------------------------------------------------------------------------------|
| No shares outstanding outside the US                                                                  | FS07 size and the FS05 index benchmark use traded value as a proxy                         | Medium | Medium      | Shares-outstanding history or free-float index weights   | none                                                                                  |
| Short history (from 2012)                                                                             | FS06 seasonality averages 1–13 years (paper: 20); FS11 starts 2014; holdout only 6.7 years | Medium | Medium      | Extend back with ART (1996–2013) for EU, UK, DK, US      | ART PIT database                                                                      |
| Oslo missing from SCANDI; DK has ~19 names                                                            | SCANDI and DK sorts use 3–5 groups of 6–12 stocks; noisy                                   | Medium | Low         | Add OBX history; report DK only as a robustness market   | none                                                                                  |
| US prices without high/low or volume in our Sharadar extract                                          | FS01 true-range N; US size-independent liquidity signals                                   | Low    | Low         | Re-pull Sharadar SEP with OHLCV                          | Sharadar subscription                                                                 |
| Event and positioning data not used: earnings dates, analyst revisions, short interest, options, news | 360° gaps in FS02 (news vs no-news jumps, implied skew, borrow)                            | Medium | Medium–High | Check coverage of the Project2 caches, start with the US | Project2 pit_short_positions, pit_options, pit_headlines, pit_13f; ART IBES (to 2013) |

### 3.3 Model

| Gap                                                                         | Where it bites                                                                                          | Impact | Effort | Remedy                                                                   | Source we already have          |
|-----------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------|--------|--------|--------------------------------------------------------------------------|---------------------------------|
| Sort-based tests only; no multivariate (Fama-MacBeth) regressions           | overlapping signals (momentum family, low-risk family) are judged one at a time                         | High   | Medium | Monthly cross-sectional regressions on all library signals, per universe | code/signals*.py library        |
| Raw 252-day betas, no shrinkage                                             | FS10 BAB, beta-neutral fixes, attribution                                                               | Medium | Low    | Vasicek or Frazzini-Pedersen shrinkage                                   | —                               |
| Residual momentum on a market-only model, 24 months                         | FS11                                                                                                    | Low    | Low    | Fama-French 3-factor residuals (US), 36 months when history allows       | French factors (Project2 cache) |
| No factor model for Europe or SCANDI inside JKP; French Europe used, in USD | attribution alphas for EU and SCANDI                                                                    | Medium | Low    | Build local-currency factor returns from the library itself              | library L/S returns             |
| Equal weighting and one-month holding everywhere                            | turnover-heavy strategies (FS08, FS06); comparability with papers that value-weight or hold 6–12 months | Medium | Low    | Add value-weighted (US) and overlapping 3/6/12-month holdings            | —                               |

### 3.4 Implementation

| Gap                                                                          | Where it bites                                        | Impact | Effort | Remedy                                                            | Source we already have                                 |
|------------------------------------------------------------------------------|-------------------------------------------------------|--------|--------|-------------------------------------------------------------------|--------------------------------------------------------|
| Flat 10 bp per side; no size- or liquidity-dependent costs, no market impact | all net numbers; small names in DK/SCANDI understated | High   | Medium | Spread and impact model by traded value (e.g. square-root impact) | Project1 art_cost_model, Saxo measured fees (Project2) |
| No taxes, dividend withholding or currency hedging costs                     | Danish and cross-border investors                     | Medium | Medium | Investor-specific net-of-tax layer                                | Project1 art_measured_taxes                            |
| Close-to-close execution, monthly rebalance at month-end                     | all strategies; month-end crowding                    | Low    | Low    | Next-day VWAP proxy; staggered rebalance days                     | —                                                      |

### 3.5 Coverage

| Gap                                                                                                          | Where it bites                              | Impact | Effort                      | Remedy                                                               | Source we already have        |
|--------------------------------------------------------------------------------------------------------------|---------------------------------------------|--------|-----------------------------|----------------------------------------------------------------------|-------------------------------|
| Missing factor families: value, quality, investment, earnings momentum, analyst revisions, payout, liquidity | the multifactor battery would be price-only | High   | Medium (US) / High (non-US) | US stock-level from Sharadar now; JKP factor-level for UK, DK, World | Sharadar US fundamentals; JKP |

### 3.6 Validation

| Gap                                                                                                                           | Where it bites                                                                             | Impact | Effort | Remedy                                                                                                                              | Source we already have             |
|-------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------|--------|--------|-------------------------------------------------------------------------------------------------------------------------------------|------------------------------------|
| The six universes overlap and co-move: World contains US, UK and EU; EU contains the Danish, Swedish and Finnish blue chips   | every 'positive in k of six' count; six markets are worth fewer than two independent tests | High   | Low    | Report the effective number of markets; judge on pooled evidence; treat World as a summary, not a seventh market; run EU ex-Nordics | the L/S series already on disk     |
| No sector or industry neutrality                                                                                              | low volatility and BAB may be utilities/staples bets; momentum partly industry momentum    | Medium | Medium | Sector-neutral sorts (US from Sharadar sectors; elsewhere ICB/GICS from index files)                                                | Sharadar TICKERS sector field (US) |
| No stress tests of named episodes (2015–16, COVID crash and rebound, 2022 rate shock) and no crash-risk analysis for momentum | FS03, FS09, FS11 (momentum crashes); FS04, FS10 (rate sensitivity)                         | Medium | Low    | Episode table and Daniel–Moskowitz bear-market/rebound regression                                                                   | —                                  |
| No capacity estimate                                                                                                          | which books survive at realistic AUM, above all DK and SCANDI                              | Medium | Medium | Capacity at a participation cap (e.g. 5% of daily traded value) per book                                                            | traded-value panels (PIT)          |
| No unit tests of the engine (look-ahead, signal lags, cost accounting)                                                        | all factsheets                                                                             | Medium | Low    | Synthetic-data tests: random signals must give ~0 before costs; a shifted signal must not change results                            | —                                  |

### 3.7 Process

| Gap                                               | Where it bites | Impact | Effort | Remedy                             | Source we already have |
|---------------------------------------------------|----------------|--------|--------|------------------------------------|------------------------|
| Repo assembly script does not know the factsheets | publishing     | Low    | Low    | Add factsheets to assemble_repo.py | —                      |

## 4. Recommended next steps

1. **Close the remaining cheap gaps:** a size- and liquidity-dependent cost and impact model, engine unit tests on synthetic data, named-episode stress tests and sector-neutral variants of the low-risk and momentum books.
2. **Add the missing families in the US at stock level** from the Sharadar fundamentals already on disk (value, quality/profitability, investment, payout, earnings momentum), and at factor level for UK, Denmark and World from JKP.
3. **Run a multivariate test** (monthly Fama-MacBeth regressions of returns on all library signals, per universe) before the battery filter, so substitutes are recognised as such.
4. **Then the multifactor battery** as planned: pre-registered filters on net return, Sharpe and turnover per segment; rolling 3-year correlations; dynamic weights over 12/36/60-month windows with equal, linear and exponential decay and volatility scaling; strict walk-forward; deflated Sharpe on the final pick.
5. **Longer history** from the ART archive (1996–2013) for EU, UK, DK and US, to give every conclusion a genuine pre-2013 out-of-sample check.

## 5. Reproduce

code/build\_overview.py reads results/xs\_\*.json, turtle\_summary.json, lottery\_summary.json, fix\_summary.json, monkey\_summary.json and the neighbourhood files.